

CFD Validation of Engine Inlet Duct Pressure Loss Model for Subsonic Non-Uniform Cross-Section Duct

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The aerodynamic performance of the flow inside the duct section following the cowl lip station directly affects the efficiency of the aircraft engine inlet. There are already various studies about pressure loss in parallel rectangular and conical ducts documented. However, a general model explaining pressure recovery within a non-uniform cross-section duct still needs research and validations. The classic book by Seddon, *Intake Aerodynamics*, introduces a prediction model for duct pressure loss, and its experimental validations are shown for conical diffusers. However, no such experimental validations or CFD simulations can be seen for other ducts of non-uniform cross-section complex geometries. This paper will start dealing with the computation of the duct loss model for ducts of non-uniform cross-sections using the numerical integration method. Then, the developed CFD validations using ANSYS Fluent for predicted pressure recovery ratio in two geometries are presented. This paper will hopefully provide the industry with a simple but reliable calculation method, reducing the reliance on time-consuming and expensive CFD simulation.

I. Introduction

The aircraft engine inlet is one of the most essential parts in high-speed aircraft and propulsion system designs because it controls the airflow into the engine. Several types of common engine inlets include two-dimensional inlet, serpentine inlet and bellmouth inlet. A two-dimensional inlet properly compresses and guides air (usually supersonic air) to the engine; it plays a vital role in maintaining average performance and efficiency of the engine at high speed [1]. Serpentine inlets, featuring a series of curves and bends, are specialized for stealth aircraft, hiding the engine from direct line-of-sight to reduce radar detection [2]. Bellmouth inlets, with a wide, bell-shaped opening, are typically used in slower aircraft or stationary testing environments, providing a smooth, unrestricted airflow to the engine [3]. Innovative designs and materials are researched and developed to enhance the performance of these inlets while meeting the challenges of shockwave control, pressure recovery, and air distortion needed for successfully designing and developing advanced-performing supersonic vehicles [4]. This massive level of importance and interests bring forward the critical necessity of this particular aircraft engine component design in setting new frontiers for aerospace technology.

The aerodynamic duct, located just after the cowl lip station, is a critical element in the design of high-speed aircraft inlets. This section plays a crucial role in maintaining the engine performance under varying flow speeds. In supersonic conditions, oblique shocks are formed within this duct, effectively compressing the incoming airflow, which is a process vital for maintaining the engine's efficiency at high speeds [5]. In low-speed or subsonic scenarios, the dynamics within the duct are significantly influenced by viscous effects [6]. Under these conditions, the airflow is susceptible to turbulence, resulting in total pressure losses. This article shows a complete Computational Fluid Dynamics (CFD) validation using ANSYS Fluent of the duct pressure loss model introduced in the seminal textbook by Seddon, *Intake Aerodynamics*[7]. The validations include two types of non-uniform cross-section ducts, non-parallel rectangular ducts and bellmouth inlet ducts, which represent how well Seddon's model can be applied under different and complex aerodynamic condition specifications.

II. Motivation

Numerous methods have been developed for the calculation of pressure loss in airflow pipes and ducts within academic research, showing the depth and diversity of theoretical and computational approaches in this branch of fluid dynamics. The Darcy-Weisbach equation is one of the most fundamental equations in fluid mechanics, used to estimate

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frictional pressure or head loss in a pipe or duct through which a fluid flow is passed. It relates this loss to pipe length, fluid velocity, density, viscosity, and the Darcy friction factor, a dimensionless coefficient dependent on the pipe's characteristics and the flow regime[8]. The equation is shown below:

$$\frac{\Delta P}{L} = f_D \cdot \frac{\rho}{2} \cdot \frac{v^2}{D} \quad (1)$$

Where:

- ρ : the density of the fluid
- D : the hydraulic diameter of the duct. $D = 2ab/(a + b)$
- v : the mean flow velocity
- f_D : the Darcy friction factor

However, the Darcy-Weisbach equation is particularly well-suited for airflow pipes with a constant cross-sectional area, as the hydraulic diameter remains constant in the equation.

Another famous method for calculating duct pressure recovery is the equation detailed in *Intake Aerodynamics*, a seminal work in the field. The equation is presented as follows (This equation could be used for both approach loss and duct loss calculation, in this paper only the duct loss will be focused on):

$$\frac{\Delta P}{q_1} = \int_1^2 \frac{q}{q_1} C_f \frac{g}{A} dl = \int_1^2 \left(\frac{A_1}{A}\right)^2 C_f \frac{g}{A} dl \quad (2)$$

Where:

- dl : length integral variable
- A_1 : entry area
- A : integral area variable
- q_1 : dynamic pressure at entry area
- C_f : skin friction coefficient
- g : perimeter of the cross section

This equation provides a method of predicting total pressure loss in stream tubes with varying cross-sectional areas bounded fully or partially by a solid surface. The underlying assumption of this equation is instantaneous mixing, which means there are uniform pressure and velocity conditions at each engine station. However, its validation in *Intake Aerodynamics* is limited to conical diffusers of varying cone angles.[9], lacks explicit validation for various types of non-uniform cross-section ducts. This limitation highlights a need for further computational or experimental validations to extend equation's applicability to other complex geometries. This paper starts with the computation of the total pressure loss for ducts featuring non-uniform cross-sections, employing trapezoidal numerical integration method. Subsequently, the paper presents CFD validations of the equation's application to two types of engine inlet ducts: non-parallel rectangular ducts and bellmouth inlet ducts, as shown in Fig. 1 and Fig. 2.



Fig. 1 Rectangular duct[10]

III. Methods

A. Numerical Calculation of Duct Pressure Loss Model

The method presented in this paper has been implemented in Python and the source code is publicly accessible, offering the research community an opportunity to review, utilize, and build upon the existing work. This model is



Fig. 2 Bellmouth duct[11]

presently developed for subsonic turbulent flow conditions within non-uniform cross-section ducts in order to capture the viscous effect. The algorithm of this model is shown below:

1. Calculate the Duct Integral

The duct pressure loss model equation (Eq. 2) could also be expressed as:

$$\frac{\Delta P}{q_1} = \int_1^2 \left(\frac{A_1}{A}\right)^2 C_f \frac{g}{A} dl = IC_f \quad (3)$$

Where I is the *duct integral*. To calculate this integral, the steps are listed below:

- 1) Calculate the area at specific x location. For the non-uniform rectangular duct, the area is calculated using linear interpolation:

$$A = A_{cowl} - (A_{cowl} - A_{th}) \cdot \frac{x}{x_{End}} \quad (4)$$

Where:

- A_{cowl} : cowl lip station area
- A_{th} : throat area
- x : specific x location inside the duct
- x_{End} : the end location inside the duct
- A : area at specific x location

For bellmouthduct, the area is calculated using the duct radius at specific x location. The radius could be obtained from the bellmouth guide lane profile:

$$A = \pi r(x)^2 \quad (5)$$

- 2) Calculate the perimeter of the cross section. For the non-uniform rectangular duct:

$$g_{peri} = (A/b + b) \cdot 2 \quad (6)$$

Where:

- b : width of the cross section
- g_{peri} : perimeter of the cross section

For the bellmouth duct:

$$g_{peri} = 2\pi r(x) \quad (7)$$

- 3) Calculate the sonic throat area using the function:

$$\frac{A_{cowl}}{A^*} = \frac{1}{M_{cowl}} \left(\frac{1 + \frac{\gamma-1}{2} M_{cowl}^2}{\frac{\gamma+1}{2}} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad (8)$$

Where:

- M_{cowl} : Mach number at cowl lip station area
- γ : ratio of specific heats
- A^* : sonic throat area

4) Using the area ratio to get the Mach number at specific x location:

$$\frac{A}{A^*} = \frac{1}{M} \left(\frac{1 + \frac{\gamma-1}{2} M^2}{\frac{\gamma+1}{2}} \right)^{\frac{\gamma+1}{2(\gamma-1)}} \quad (9)$$

In this function, Mach number is assumed to be a symbolic variable, and python package *sympy* is used to solve the equation. The final output of this function will be a subsonic Mach number corresponding to the area ratio.

5) Calculate the density at this x location using the relations:

$$\frac{\rho_{cowl,0}}{\rho_{cowl}} = \left(1 + \frac{\gamma-1}{2} M_{cowl}^2 \right)^{\frac{1}{\gamma-1}} \quad (10)$$

$$\frac{\rho_{cowl,0}}{\rho} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{1}{\gamma-1}} \quad (11)$$

Here:

- $\rho_{cowl,0}$: the stagnation density before entering the duct.
- ρ_{cowl} : the static density before entering the duct.

6) Calculate the temperature at this x location using the relations:

$$\frac{T_{cowl,0}}{T_{cowl}} = 1 + \frac{\gamma-1}{2} M_{cowl}^2 \quad (12)$$

$$\frac{T_{cowl,0}}{T} = 1 + \frac{\gamma-1}{2} M^2 \quad (13)$$

Here:

- T_{cowl} : the static temperature before entering the duct.
- $T_{cowl,0}$: the stagnation temperature before entering the duct.

7) Calculate the speed of sound and local velocity:

$$SoS = \sqrt{\gamma RT} \quad (14)$$

$$U = M \cdot SoS \quad (15)$$

8) Calculate the hydraulic diameter. For the non-uniform rectangular duct:

$$D_h = 2 \cdot \frac{A}{A/b + b} \quad (16)$$

For bellmouth duct, the diameter is simply:

$$D_h = 2r(x) \quad (17)$$

9) Calculate the Reynolds number at specific x location:

$$Re_x = \frac{\rho U D_h}{\mu} \quad (18)$$

10) Calculate the skin friction coefficient. In this model, standard correlation for a fully turbulent boundary layer over a smooth flat plate is employed [12]:

$$C_f = 0.0576 Re_x^{-0.2} \quad (19)$$

11) Finally calculate the duct integral:

$$I(x) = \left(\frac{A_{cowl}}{A} \right)^2 \frac{g_{peri}}{A} \quad (20)$$

2. Numerical Integration Method Implementation

The trapezoid method is a numerical approach for estimating the definite integral of a function by dividing the area under the curve into trapezoids. It calculates the area of each trapezoid and sums them up for the approximation. This method improves accuracy by increasing the number of trapezoids, and is effective for smoothly varying functions [13]. The algorithm of this step includes:

1) Calculate the dynamic pressure using the relation:

$$q_{Cowl} = \frac{1}{2} \rho_{Cowl} U_{Cowl}^2 \quad (21)$$

2) Divide the x direction length into n intervals, then calculate the unit interval length (dx).

3) Initiate the output expression (temporary) as:

$$\left(\frac{\Delta P}{q_1} \right)_{temp} = 0.5(I(x=0) + I(x=x_{End})) \quad (22)$$

4) Go through the for loop for all the intervals, accumulate the result:

$$\left(\frac{\Delta P}{q_1} \right)_{temp,new} = I(x=x_n) + \left(\frac{\Delta P}{q_1} \right)_{temp,old} \quad (23)$$

5) Finalize the result by multiplying dx :

$$\left(\frac{\Delta P}{q_1} \right)_{final} = \left(\frac{\Delta P}{q_1} \right)_{temp} \cdot dx \quad (24)$$

6) Calculate the pressure recovery ratio:

$$PR = 1 - \Delta P / P_{t,cowl} \quad (25)$$

B. Computer-Aided Design (CAD) Model

1. Non-Parallel Rectangular Duct

A non-parallel rectangular duct CAD model is generated by NX. The cowl lip and throat cross sections are shown in Fig. 3.

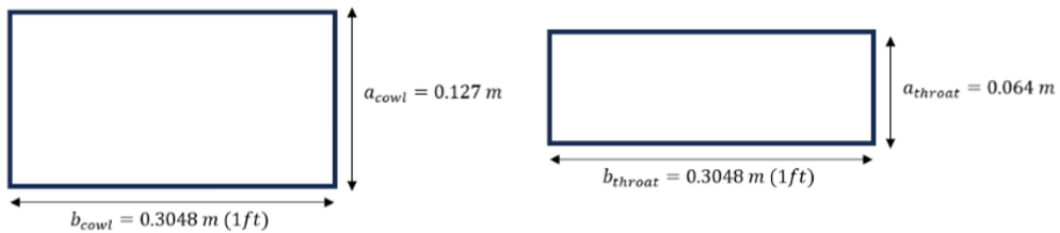


Fig. 3 Cross sections of the rectangular duct

The side and isometric view of the duct is shown below:

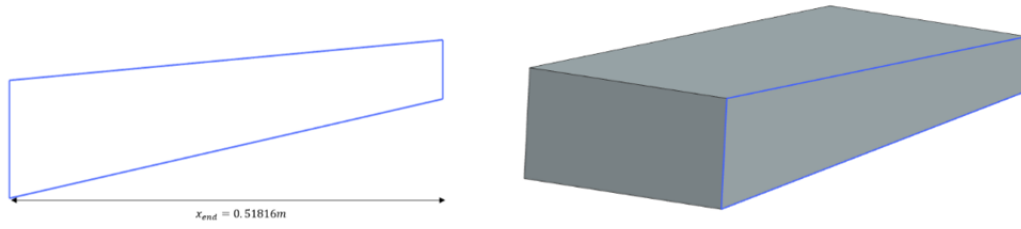


Fig. 4 Side and isometric view of the rectangular duct

The important dimensions of the non-uniform rectangular duct are shown below:

Table 1 Dimensions of non-uniform rectangular duct

Notations	Dimensions	Values
b_{cowl}	Cowl lip section width	0.3048 m
a_{cowl}	Cowl lip section height	0.127 m
b_{throat}	Throat section width	0.3048 m
a_{throat}	Throat section height	0.064 m
x_{end}	Throat section x location	0.51816 m

2. Bellmouth inlet duct

A bellmouth inlet duct CAD model is also generated by NX. The cowl lip and throat cross sections are shown below:

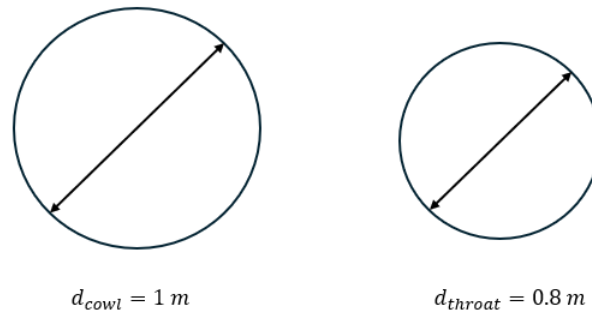


Fig. 5 Cross sections of the bellmouth duct

The side and isometric view of the duct is shown in Fig.6, and the important dimensions of the bellmouth duct are shown in Table 2.

C. Fluent CFD Simulation Setup

1. Non-parallel rectangular duct

In this project, Fluent is utilized to conduct the CFD simulation of the low speed turbulent flow passing through the non-parallel rectangular duct. The generated CAD file is fluid-only region, and the surface mesh minimum size is 0.5mm, maximum size is 12mm. For the boundary type of each surface, the inlet surface is set as *velocity-inlet*, the

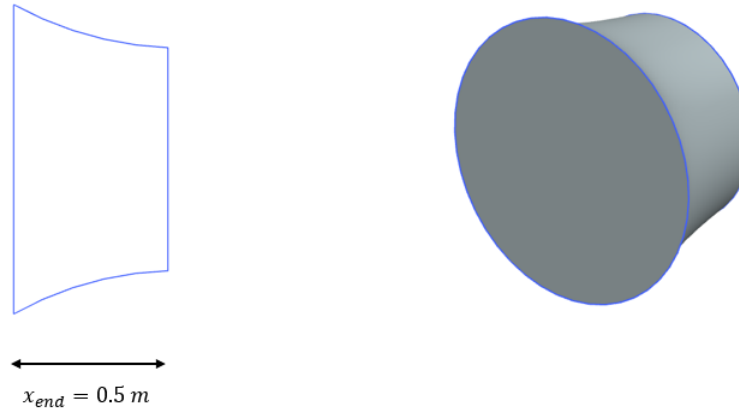


Fig. 6 Side and isometric view of the bellmouth duct

Table 2 Dimensions of bellmouth duct

Notations	Dimensions	Values
r_{cowl}	Cowl lip section radius	0.5 m
r_{throat}	Throat section radius	0.4 m
x_{end}	Throat section x location	0.5 m

outlet surface is set as *outflow*. All other surfaces are set as *wall*. The boundary layers addition is based on Sutherland law and setting y^+ as 1. Smooth transition offset method is chosen, with 28 boundary layers and growth rate as 1.2. The volume mesh is filled with tetrahedral, also with growth rate as 1.2. The final mesh visualization is shown below:

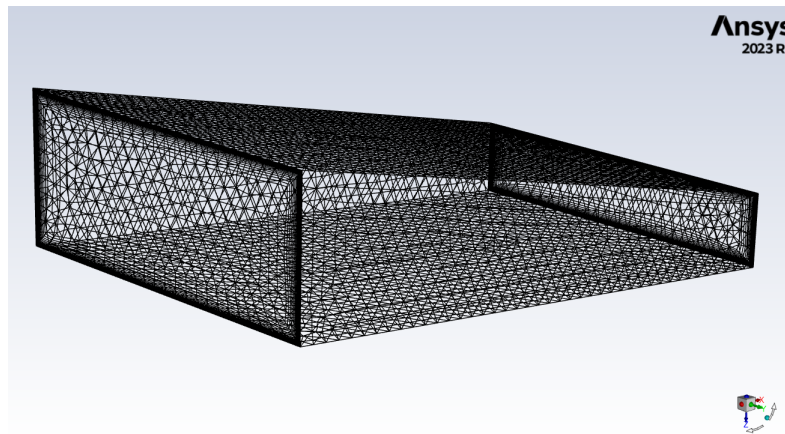


Fig. 7 Rectangular duct mesh visualization

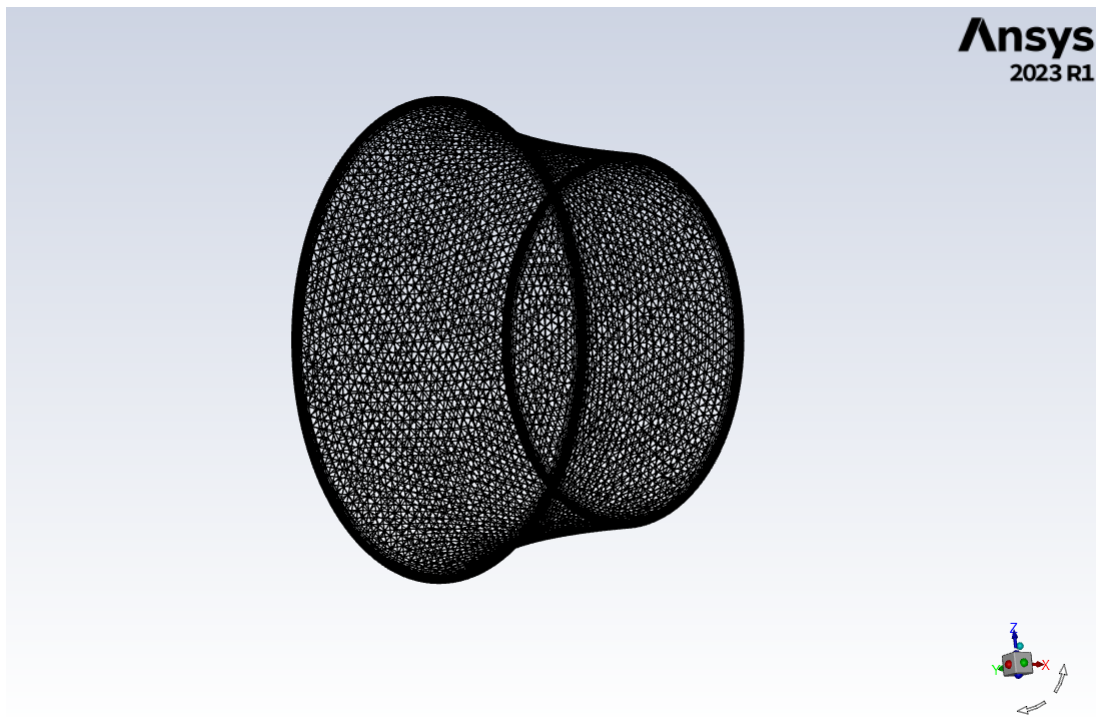
Then the meshing is imported into Fluent solution interface. Some important initial conditions are shown in Table 3. The shear stress transport (SST) $k - \omega$ model [14] is utilized for CFD simulation. SST model is a combination of $k - \omega$ and $k - \epsilon$ models with the use of $k - \omega$ in the inner parts of the boundary layer, which is suitable for Low-Re turbulence cases without any extra damping functions. In the area close to free-stream area, SST model is switched to $k - \epsilon$ model, avoid $k - \omega$ model's sensitivity to the inlet free-stream turbulence properties.

Table 3 Initial conditions for rectangular duct

Variables	Values
Far field flow pressure	101325 Pa
Far field flow temperature	288 K
Far field flow density	1.225 kg/m^3
Viscosity	$1.81\text{e-}5 \text{ kg/(m} \cdot \text{s)}$
Ratio of specific heats	1.4
Inlet velocity	Depends on the Mach number
Air properties	Real gas
Number of iterations	250

2. Bellmouth inlet duct

The meshing, solver settings and the initial conditions are the same as those of rectangular duct. The final mesh visualization is shown below:

**Fig. 8 Bellmouth duct mesh visualization**

IV. Results

A. Non-parallel rectangular duct

After the CFD simulation, the total pressure contour of rectangular duct in Fluent is visualized in Fig. 9. Notice that in the visualization the total pressure is gauge pressure, respect to the reference pressure, and it is representing the point pressure. The surface integral function is utilized to obtain the area-weighted total pressure at the cowl lip section and throat section. Using the difference of the total pressure between cowl lip and throat surface, divide it by the total pressure at the cowl lip station to get the pressure loss ratio. Repeat all these steps for other Mach numbers.

In this paper, the chosen maximum Mach number is 0.3. For the flow with low Reynolds number, the viscous forces are dominant. But when Reynolds number gets higher, the inertia forces will dominate[15]. The proposed model is intended to capture the viscous effect at low Reynolds number condition, so the Mach number is set to be low. The comparison between numerical prediction results and CFD simulation results is shown in Table 4.

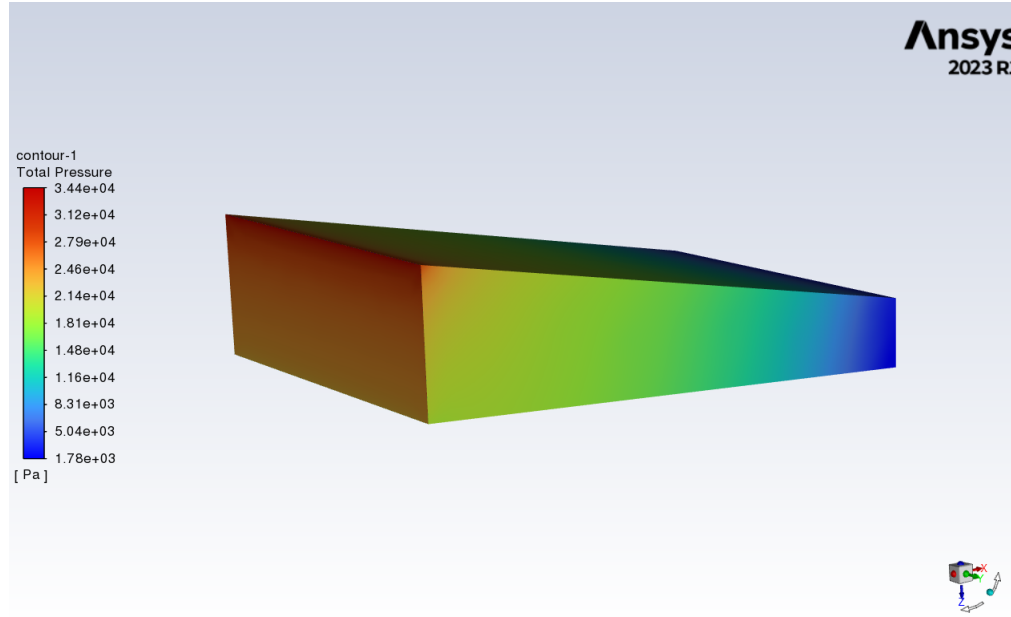


Fig. 9 Rectangular duct total pressure contour

Table 4 Rectangular duct pressure recovery ratio comparisons

Mach	Duct Pressure Recovery Model	CFD Results	Error Percentage
0.01	0.99998481	0.99998097	0.00039%
0.05	0.99973262	0.99968074	0.0052%
0.1	0.99907378	0.99887282	0.0201%
0.2	0.99684057	0.99581009	0.1035%
0.3	0.99366068	0.99220050	0.1472%

As demonstrated in the table, the pressure recovery model exhibits high accuracy. For the cases with Mach numbers under 0.3, the error percentages are lower than 0.15%, indicating the model's high precision. Another observation is that as the Mach number increases, the error percentage would also rise. This may be caused by the increase of Reynolds number, which will result in the viscosity becoming less dominant. The prediction model is highly depending on the viscous effect on the wall, so the error percentage would increase.

B. Bellmouth inlet duct

After the CFD simulation, the total pressure contour of bellmouth duct in Fluent is visualized in Fig. 10. The comparison between numerical prediction results and CFD simulation results is shown in Table 5. Based on the observations, the pressure recovery ratio also shows high accuracy in bellmouth inlet duct, with same trends in non-uniform rectangular duct.

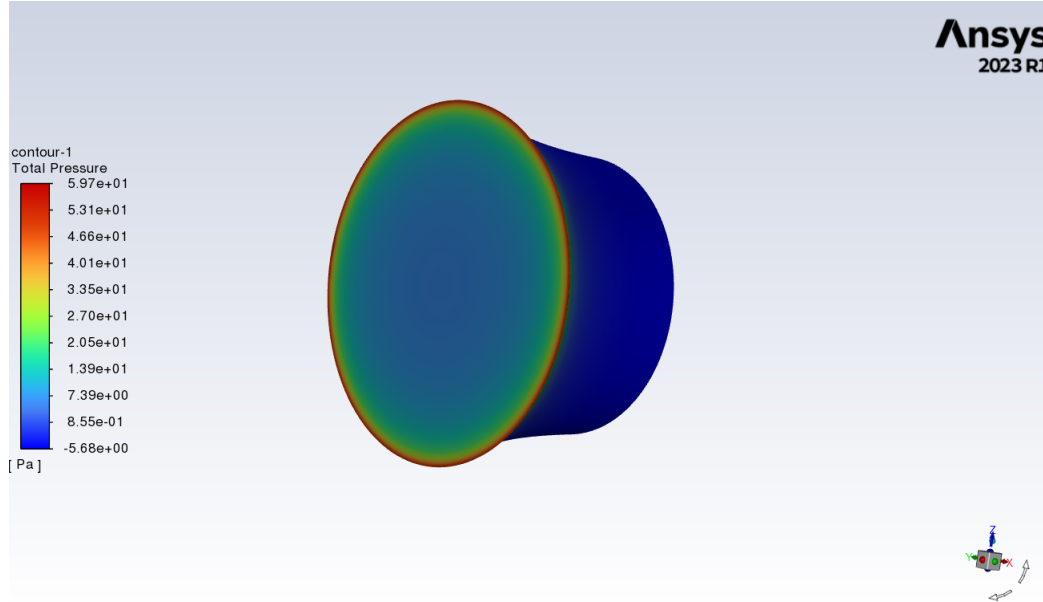


Fig. 10 Bellmouth duct total pressure contour

Table 5 Bellmouth duct pressure recovery ratio comparisons

Mach	Duct Pressure Recovery Model	CFD Results	Error Percentage
0.01	0.99999828	0.99998026	0.0018%
0.05	0.99996896	0.99945271	0.0517%
0.1	0.99989250	0.99786495	0.2032%
0.2	0.99963332	0.99148677	0.8216%
0.3	0.99926427	0.98109466	1.852%

V. Conclusion

The paper first applies the trapezoidal method to calculate duct pressure loss and develops the basic model suggested in *Intake Aerodynamics*. Afterward, CFD simulations on several non-uniform cross-section ducts, including non-parallel rectangular duct and bellmouth inlet ducts using ANSYS Fluent are conducted. Based on the comparison of obtained results from numerical integration with those from CFD simulations, it was shown that the numerical model is highly accurate. This accuracy can be applied not only to uniform cross-section ducts and conical diffusers as stated in *Intake Aerodynamics*, but also to ducts with non-uniform and varying cross-sectional areas. Therefore, this study contributes much to understanding and predicting duct pressure loss in more complex geometries, giving a robust and reliable tool for engineers and researchers working in aircraft engine inlet design areas.

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